

Deepak Mallampati

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Kent, OH - Open to relocation (Dallas, TX)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in LangChain/LangGraph-based production agent deployment, RAG systems, and end-to-end LLM evaluation pipelines. Experienced in co-building multi-step agentic workflows for enterprise healthcare AI - from pre-sales POC design through production deployment and post-sale iteration. Proven delivery track record: ~35% retrieval precision improvement, >30% hallucination reduction, ~25% agent response stability improvement, and ~30% fewer post-deployment defects. Comfortable owning the full customer technical lifecycle across data, prompts, models, evaluation, APIs, and Docker-based deployment. Seeking a Deployed Engineer role to bridge LangChain's platform and enterprise customers building production agent systems.

CORE COMPETENCIES

LangChain / LangGraph • LlamaIndex • Agentic LLM Workflows • RAG Systems • LLM Evaluation & Observability • Guardrails & Grounding • Structured Outputs • Prompt Engineering • FastAPI / Flask • Docker • Python • TypeScript / React • Vector Search / Embeddings • MLOps / LLMOps • scikit-learn / XGBoost • PyTorch / Hugging Face • CI/CD (Jenkins) • SQL / PostgreSQL

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting
USA

- Built and deployed **LangChain-based agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) - structured reasoning, tool discipline, grounding rules; improved agent response stability by ~25%.
- Designed and iterated **RAG systems** for clinical knowledge retrieval: recursive semantic chunking, transformer-based embeddings, vector search; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%; retrieval-grounded response alignment exceeded 90%.
- Built end-to-end **LLM evaluation pipelines** with guardrails, grounding rules, and audit trails; reduced hallucinations by >30%; maintained HIPAA-conscious data governance and stakeholder-facing evaluation documentation.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging and load simulation; reduced post-deployment defects by ~30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and engagement scoring; improved recall by 15-20% on high-risk categories holding precision >90%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP
Hyderabad, India

- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Optimized SQL/T-SQL stored procedures for the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Designed **Grafana** performance dashboards to track long-running queries, deadlocks, and CPU/I/O contention; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics
Hyderabad, India

- Built transformer-based **video summarization and highlight generation** system; reduced manual review time by 60-70%; achieved ~85% **alignment** with human-curated highlights.
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~30%; deployed as a **Flask API** for non-technical staff - adopted independently without training.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (5 agents: Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor duplicate detection; explainable risk scoring with audit-ready reasoning traces.

LangChain, LlamaIndex, Python, FastAPI, Docker, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation

Shipped to the Chrome Web Store. Full-stack browser extension integrating **four AI providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handling, 7-day translation cache, per-domain selector configs for 29 sites.

TypeScript, Manifest V3, Python, multi-provider LLM integration

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

FastAPI + React/TypeScript app with **intermediate structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success rate **~25-30%** over direct-prompt orchestration.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate, structured outputs

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN with unified **YOLOv8**; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives (**~25% reduction**).

YOLOv8, OpenCV, Python, computer vision

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

LLM & GenAI: LangChain, LangGraph, LlamaIndex, RAG, agentic workflows, LLM evaluation, guardrails, grounding, structured outputs, prompt engineering, embeddings, vector search

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL)

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers

Computer Vision: YOLOv8, OpenCV, ControlNet, Stable Diffusion, video analytics

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required